



CLEAN IN ACTION
GREEN FOR
APEC

CLEAN COAL POWER TECHNOLOGIES

Status, Development and Practice



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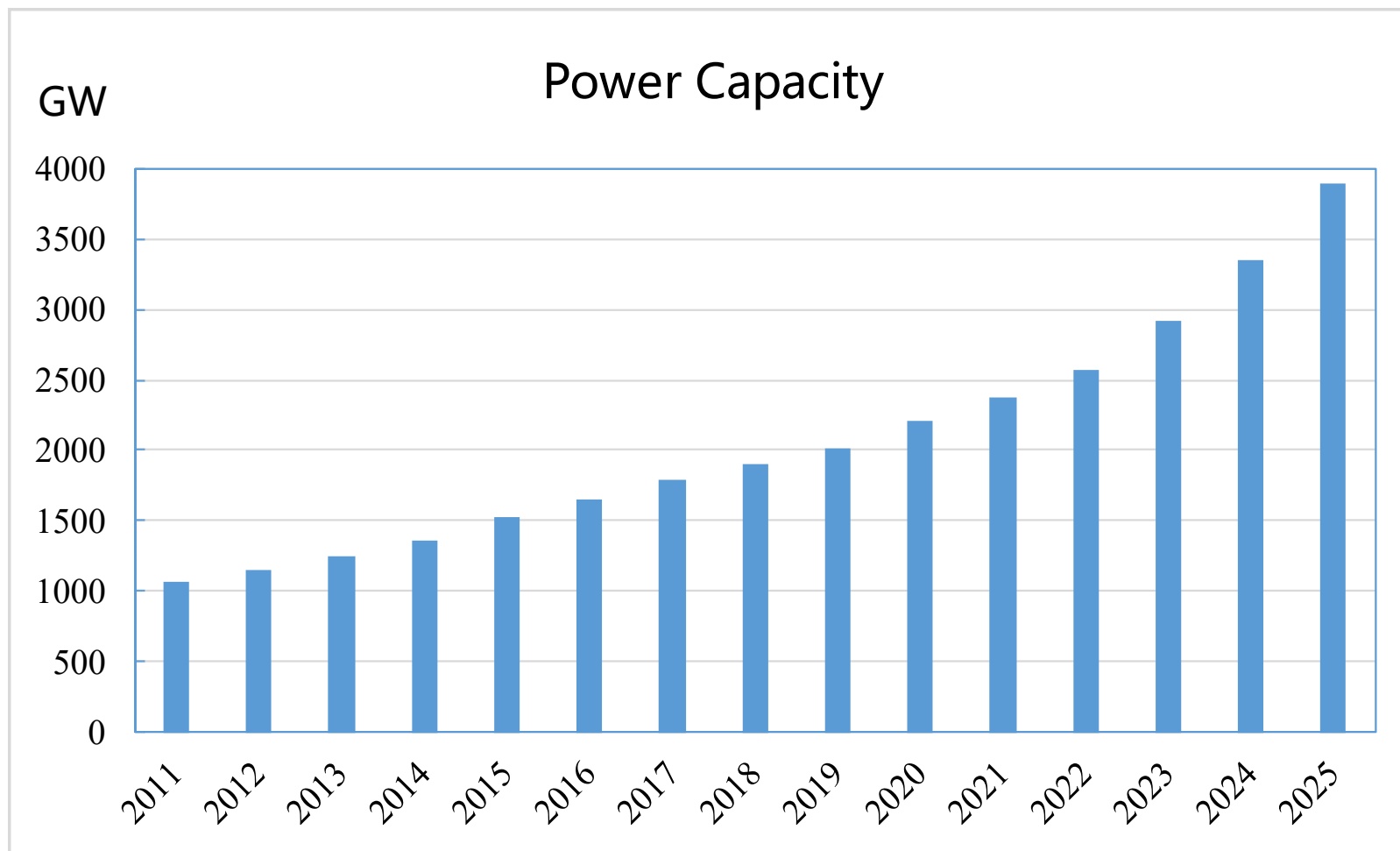
ONE

BACKGROUND OVERVIEW

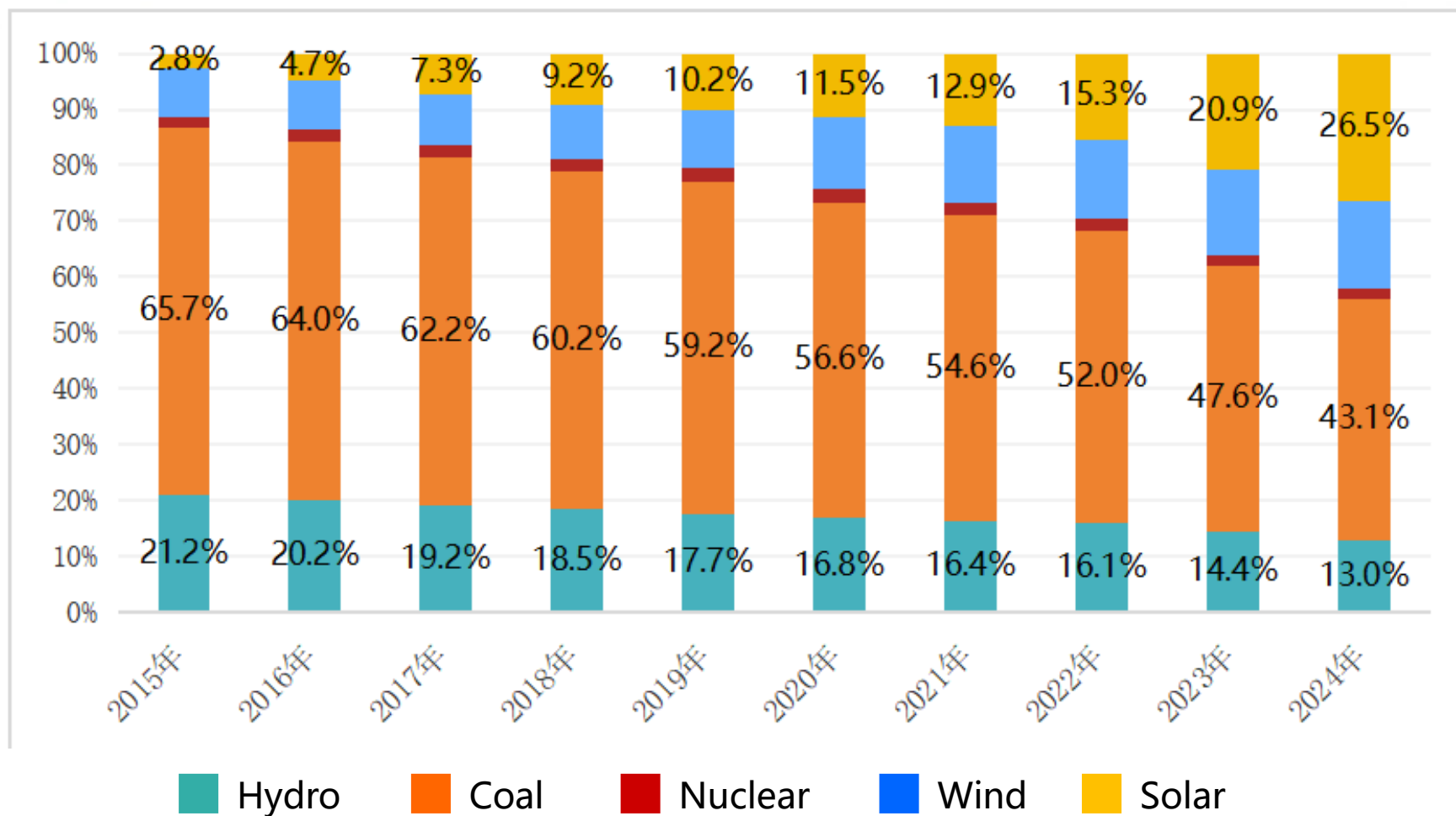
Statistics, History, Status, and Policy Guidance



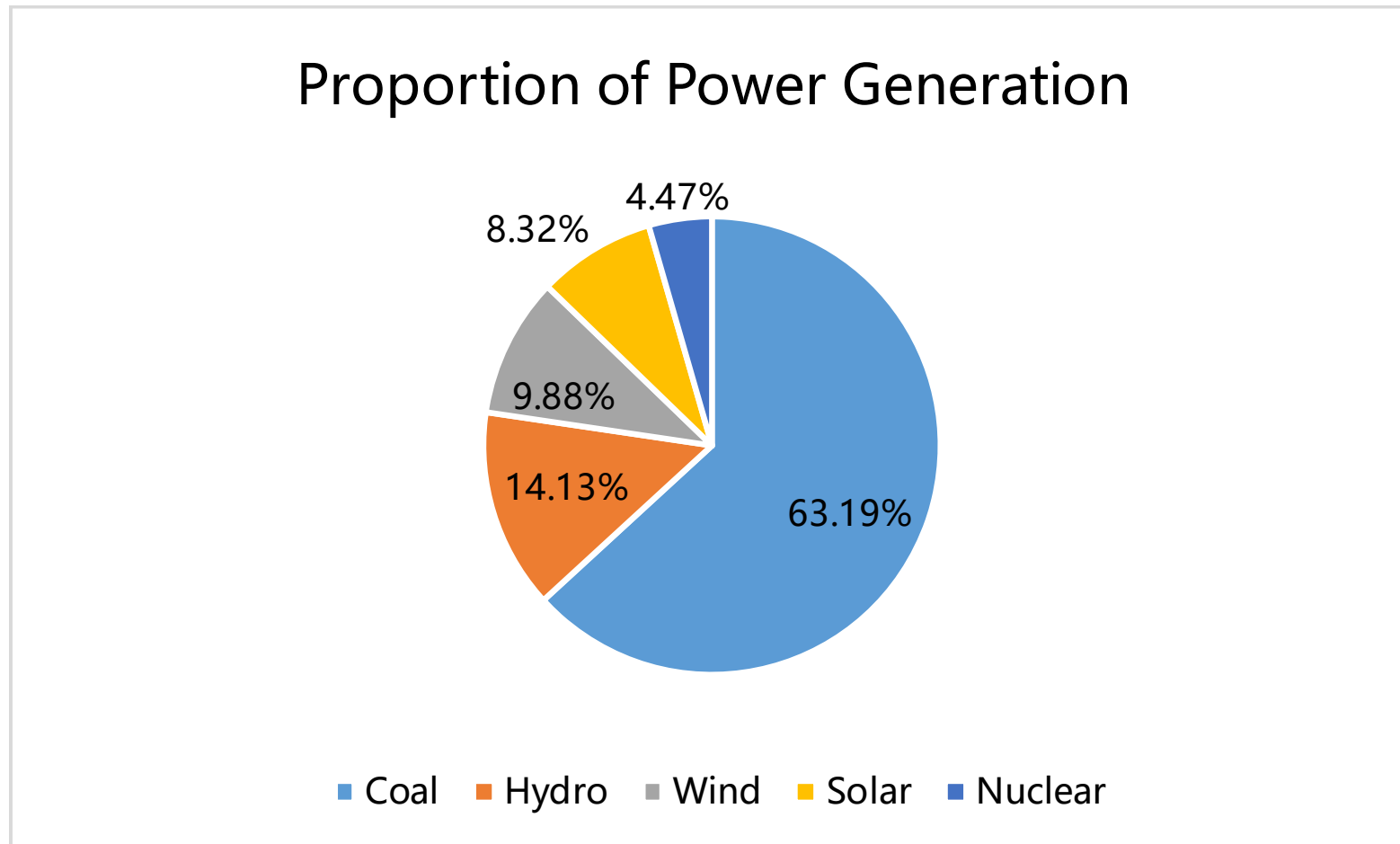
01. BACKGROUND: POWER CAPACITY OF CHINA



01. BACKGROUND: POWER CAPACITY DISTRIBUTION



01. BACKGROUND: PROPORTION OF POWER GENERATION



01. BACKGROUND: HISTORY

| HISTORICAL EVOLUTION

1. Initial Stage (1970s-1990s) | End-of-pipe Pollution Control

Focused on post-combustion Flue Gas Desulfurization (FGD) and Dust Removal Technology, to address acid rain and dust pollution, laying the foundation for environmental governance.

2. Development Stage (2000s-2010s) | Energy Efficiency & Multi-pollutant Control

Shifted towards improving power generation efficiency technologies (Supercritical Power Generation Technology) and multi-pollutant control technologies (Ultra-low Emission Technology).

3. Deepening Transformation Stage (2020s-Present) | Synergy of Pollution & Carbon Reduction

Focuses on "synergistic reduction of pollution and carbon", developing higher efficiency technologies, flexibility retrofitting, and integration with CCUS for low-carbon transition.

01. BACKGROUND: CURRENT STATUS

| CURRENT STATUS



Scale and Status: The "Ballast Stone" for Energy Security

China has built the world's largest clean coal power system, contributing 60% of power generation and 80% of peak-shaving capacity, serving as the "ballast stone" for energy security and the "stabilizer" for new energy.



Emission & Control: The "Green Benchmark" for Fossil Power

By the end of 2025, over 95% of coal-fired power units in China had achieved ultra-low emissions, reaching an internationally leading clean level.



Efficiency & Innovation: The "Power Engine" for Low-Carbon Upgrade

The average power supply coal consumption dropped to 301.6 g/kWh in 2024, with thermal efficiency exceeding 50% for the world's first 630°C Ultra-Supercritical (USC) unit.

01. BACKGROUND: POLICY GUIDANCE

| CORE POLICY GUIDANCE



"Opinions on Energy Conservation and Carbon Reduction" (April 2026)

Emphasizes "strengthening energy-saving, carbon-reducing, and flexibility retrofiting of coal power", focusing on performance improvement of existing units.



"14th Five-Year Plan for Modern Energy System"

Requires "promoting clean and efficient utilization of coal, and orderly reduction and substitution", setting a target of reducing the average power supply coal consumption to below 300 g/kWh by 2025.



"Action Plan for Carbon Peaking Before 2030"

Requires "strictly controlling new coal power projects" and vigorously promoting the "Triple Retrofit (energy-saving, CHP, flexibility retrofit" of existing units.

TWO

KEY CLEAN COAL POWER GENERATION TECHNOLOGIES



02. CORE PROCESSING STAGES FOR CLEAN COAL UTILIZATION



Pre-: Coal Washing and Coal Blending

Coal Washing

Reduces ash and sulfur through washing processes. China's raw coal washing rate reached 78.6% in 2024.

Coal Blending

Scientifically blends different coal types for stable, efficient, and low-pollution combustion.



In-: Low NO_x Burner Technology

Principle

Inhibits NO_x formation through air staging, fuel staging, and flue gas recirculation.

Effect

NO_x concentration at boiler outlet to 200-300mg/m³, reducing the burden on subsequent SCR system.



Post-: Multi-Pollutant Removal System

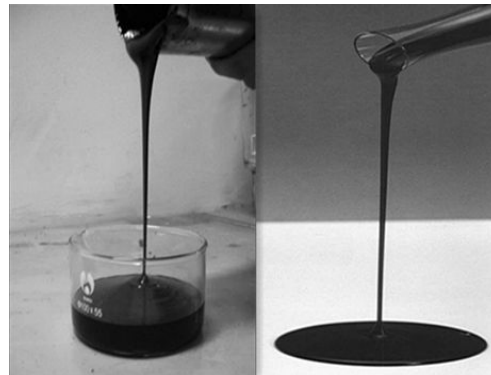
Mainstream Route

"SCR De-NO_x + WFGD De-SO₂ + High-Efficiency Dust Removal".

Core Technologies

Dust Removal (Electro-bag/WESP), Desulfurization (WFGD), Denitrification (SCR).

PRE-COMBUSTION: COAL WASHING AND COAL BLENDING TECHNOLOGIES



▲ Different Pre-combustion technologies:
coal washing, coal blending, coal briquettes, coal-water slurry

Core Principle

Remove ash and sulfur by physical or chemical coal washing to improve coal quality.
Mix different kinds of coal scientifically by coal blending to increase combustion efficiency.

Improve efficiency

Stabilize coal quality, increase boiler combustion efficiency greatly, reduce combustion fluctuations.

Clean Emissions

Remove ash and sulfur from coal source, greatly reduce dust and SO_x emissions.

Reduce Costs

Blend low-price coal, optimize fuel structure, reduce fuel costs effectively.

IN-COMBUSTION: LOW NO_x COMBUSTION TECHNOLOGY (LNB)

Air Staged Combustion

Staged supply of combustion air into the furnace creates a fuel-rich, low-oxygen reducing atmosphere in primary combustion zone, inhibiting the formation of thermal NO_x at the source.

Fuel Staged Combustion

Secondary Fuel (reburning fuel) is injected into downstream of primary combustion zone to form a strong reducing zone, which reduces the generated NO_x into harmless N₂.

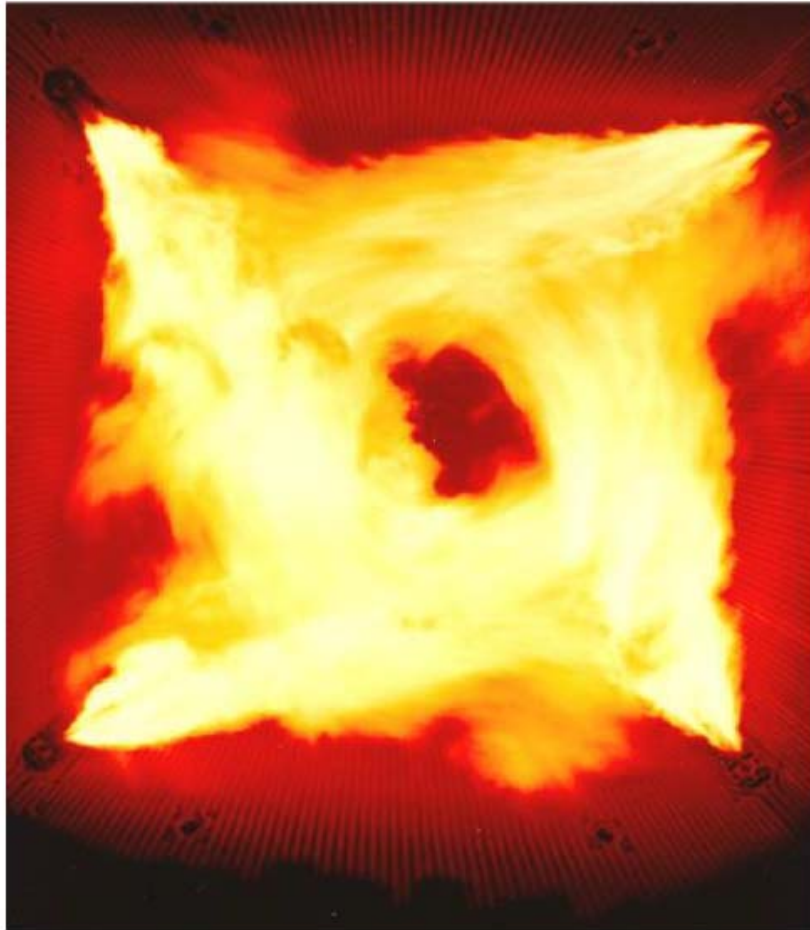
Flue Gas Recirculation (FGR)

Extracted low-temperature flue gas is mixed with combustion air to reduce O₂ concentration and flame temperature in combustion zone, effectively suppressing thermal NO_x and prompt NO_x.



Application Effect: Reduce NO_x by 50%, substantially lowering treatment load and operating costs of downstream denitrification facilities such as SCR.

LOW NO_x COMBUSTION TECHNOLOGY



Four-corner Tangential Firing



Front-and-Rear Wall Opposed Firing

CIRCULATING FLUIDIZED BED (CFB) - PRINCIPLE AND ADVANTAGES

Technical Principle: High-Efficiency Circulating Combustion System

Closed-loop system of "combustion-separation-reflux-re-combustion", which greatly improves combustion efficiency and fuel adaptability.



Wide Fuel Adaptability, Compatible with Multiple Fuel Types



Stable Operation with Wide Load, Adapting to Grid Peak Shaving

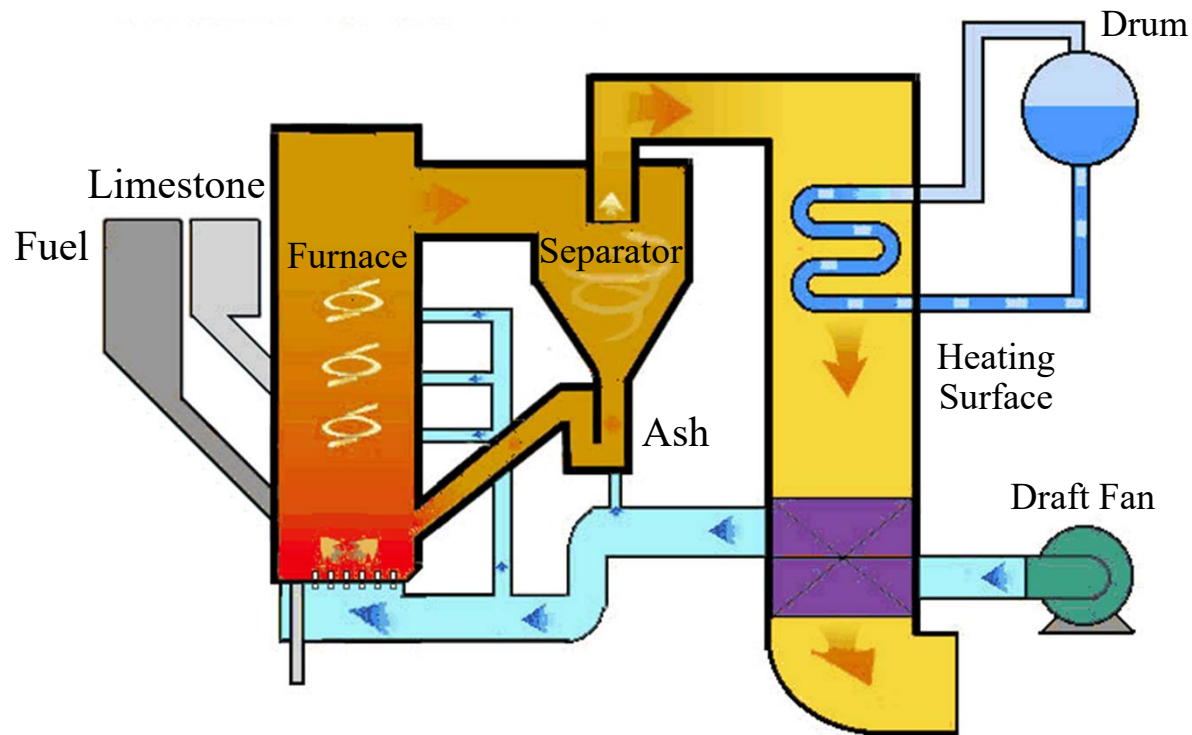


**Integration In Furnace
Desulfurization, Low Cost and High Efficiency**

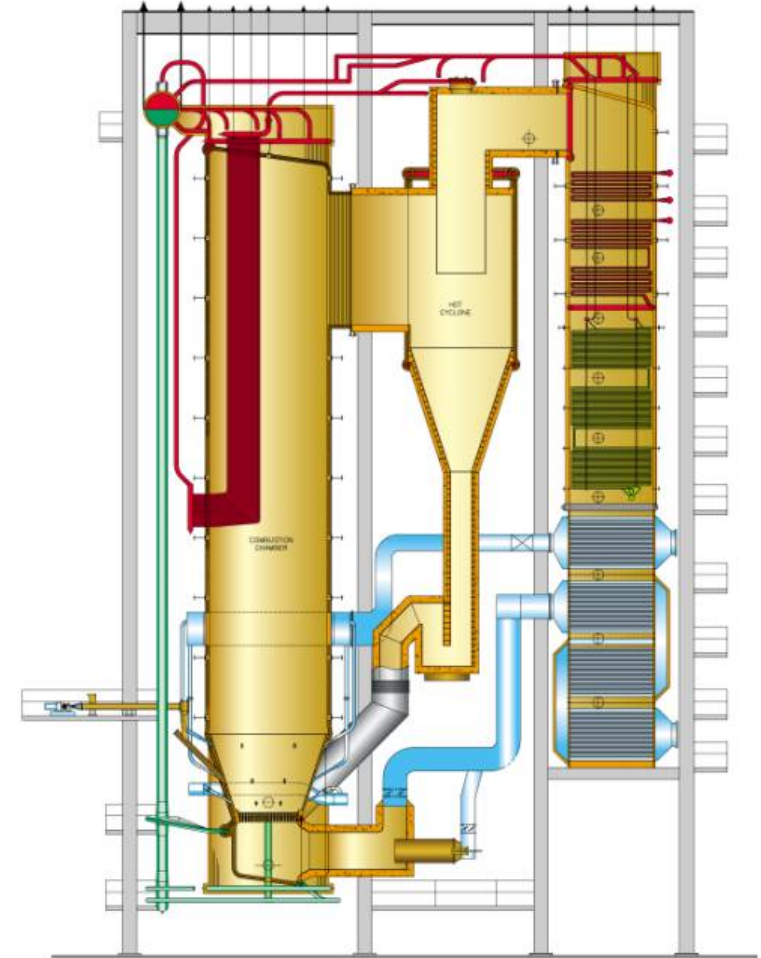


**Low-Temperature Combustion
Inhibition, Low NO_x Original Emission**

CIRCULATING FLUIDIZED BED (CFB) - PRINCIPLE



CFB Boiler Principle



CFB Boiler Structure

CIRCULATING FLUIDIZED BED (CFB) - ADVANTAGE

DIMENSION	CFB BOILER	PULVERIZED COAL BOILER
Combustion Efficiency and Power Supply Efficiency	Combustion efficiency 97.5%-98.8% ; supercritical unit power supply efficiency can reach 45% , maintaining stable conversion with low-quality coal.	Combustion efficiency > 99% ; ultra-supercritical unit efficiency > 48% , but performance is sensitive to coal quality fluctuations.
Coal Consumption and Pollutant Control	Coal consumption ~312g/kWh (low-quality coal); NO _x emission < 200mg/Nm³ . High-efficiency in-furnace desulfurization reduces tail-end facility costs.	Coal consumption ~290g/kWh (high-quality coal); NO _x emission up to 800-1200mg/Nm³ , requiring expensive tail-end purification systems.
Load Regulation and Life Cycle Cost	Load regulation 30%-110% , excellent deep peak shaving; LCOE is 0.03-0.05 CNY/kWh lower than PC boilers.	Load regulation 50%-100%, poor low-load stability; high LCOE due to reliance on high-quality coal and environmental costs.

CIRCULATING FLUIDIZED BED (CFB) - INDUSTRY

Market Position

Market Position: The world's largest manufacturer and user for CFB, accounting for over 60% market.

Technology Leadership: The world's first 660MW supercritical CFB unit has been successfully commissioned in 2020.

Industrial Achievements

CFB technology achieves dual unification of environmental and economic benefits, providing optimal technical pathway for clean and efficient utilization of high-sulfur, low-calorific coal.



Pingshuo CFB Power Plant, Shanxi Province

ULTRA (SUPER) CRITICAL POWER GENERATION - PRINCIPLE

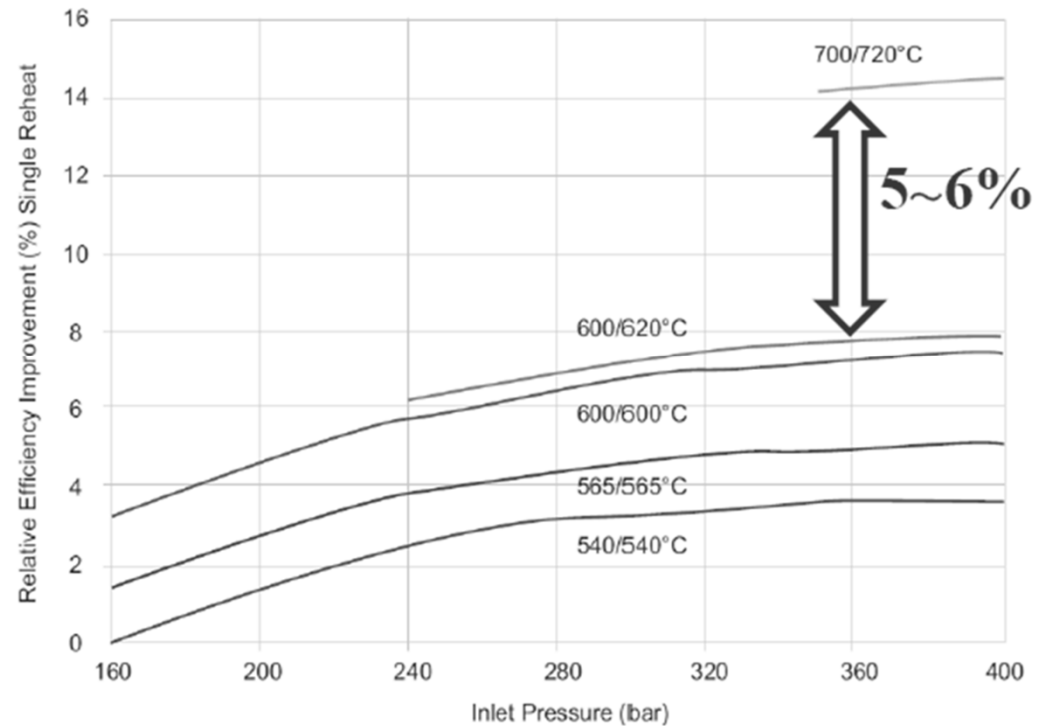
**Core Technical Principle: Breaking
Critical Point (22.12MPa, 374.15°C)**

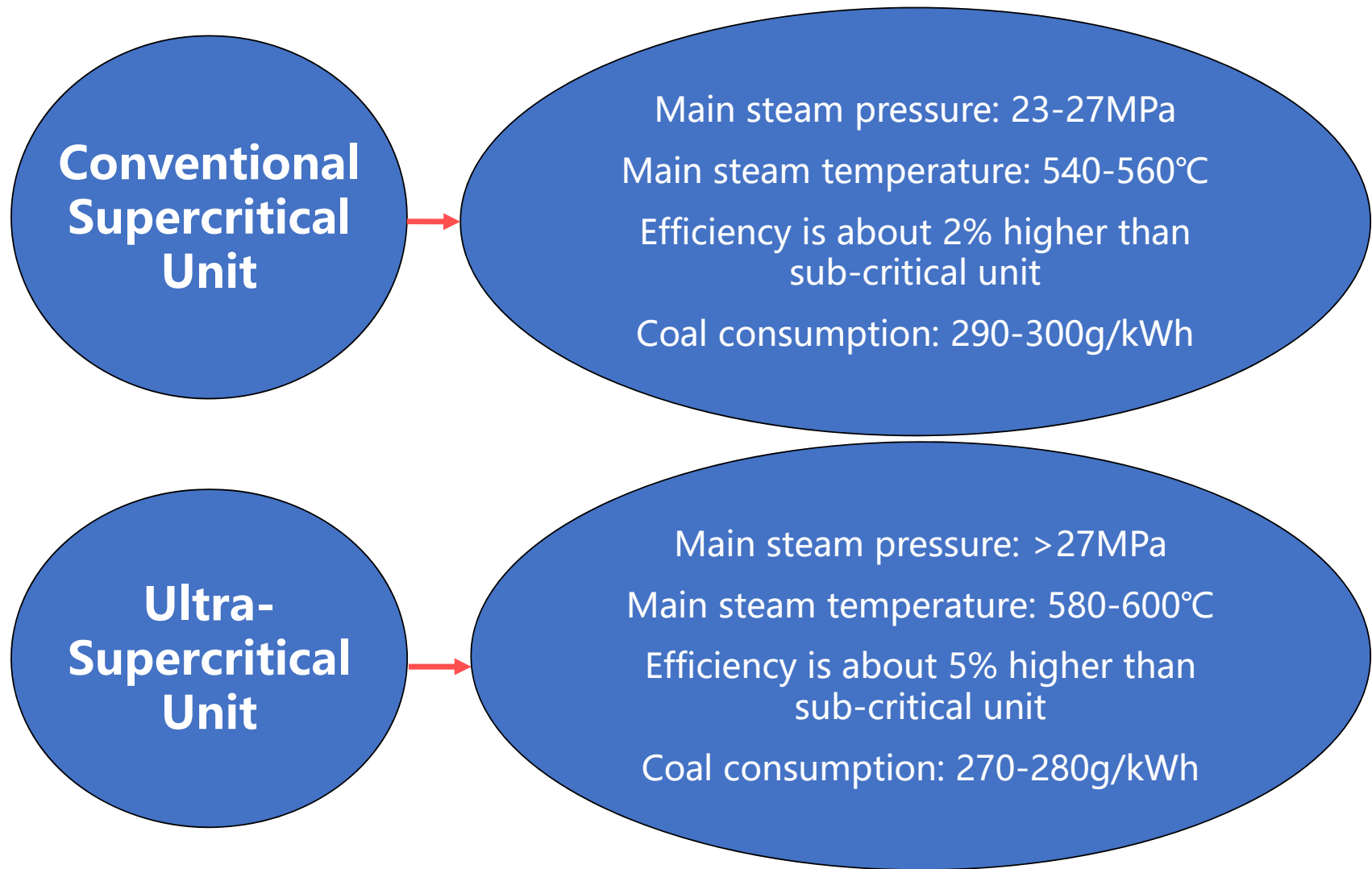


**Temperature:
Dominant Effect**



**Pressure:
Gain Effect**





ULTRA-SUPERCritical (USC) POWER GENERATION - INDUSTRY

China: Globally Leading Industrial Scale and Benchmark Projects

Largest Scale Globally: By the end of 2024, installed capacity of USC units exceeded 320GW, accounting for nearly 60% of total coal-fired power plants in China.

In June 2026, the world's first 630°C double-reheat USC million-kWt unit was commissioned.

Industry Milestone: China has changed from technological follower to global leadership in clean utilization of traditional fossil energy.



**Power Generation
Efficiency**

50.05%

**Supply Coal
Consumption**

245.7g/kWh

Source: Chinese Energy Administration and Operation Report of Datang Yuncheng 630°C double-reheat unit.

DEEP PEAK-SHAVING TECHNOLOGY - PRINCIPLE

Core Principle

Through technical retrofits, the minimum operating load limit of generating units can be overcome. Units can maintain stable combustion and safe operation under operating conditions of 20%-50% rated power, effectively improving unit flexibility and peak-shaving capacity.



Boiler-Side Retrofit

Adopt low-load stable-combustion burners and optimize furnace structure to fundamentally solve unstable combustion problem under low-load conditions.



Steam Turbine Optimization

Adopt low-pressure cylinder zero-output or cylinder-cutoff technology to greatly reduce ventilation friction loss under low-load conditions and improve efficiency.



Heat-Power Decoupling Technology

Install electrode boilers or heat storage devices to decouple heat load from electric load, so as to release more space for electric power peak regulation.



Control-System Upgrade

Optimize parameters of the coordinated control system (CCS), improving the unit's fast-response capability to power-grid load commands..

DEEP PEAK-SHAVING TECHNOLOGY - INDUSTRY



Industry Retrofit Status and Targets

By the end of 2024, China had completed flexibility retrofits for coal-fired generating units with a total capacity of 210 GW, which has greatly enhanced the power-system peak-shaving capability.

210 GW

Policy Orientation

Implement the requirement of Retrofit all eligible units, promote coal-fired power from main power source to regulating power source, and support achievement of Dual-Carbon Goals.

THREE

ADVANCED CLEAN COAL POWER TECHNOLOGIES



FLUE GAS ULTRA-LOW EMISSION TECHNOLOGY

Flue gas ultra-low emission actualize the emission concentrations of pollutants such as dust, SO₂, and NO_x are reduced to the strictest standards, achieving clean and efficient power generation.

**1. Source Control:
Low NO_x Burner (LNB)
Technology**

**2. Core Denitrification:
Selective Catalytic
Reduction (SCR)**

**3. Particle Removal:
High-Efficiency Dust
Removal Technology**

**4. Desulfurization Synergy:
Wet Flue Gas Desulfurization (WFGD)
Device**

**5. Advanced Purification:
Wet Electrostatic Precipitator (WESP)**

ULTRA-LOW EMISSION - REDUCTION STANDARD AND COST

Pollutants	2004 Standard	2012 Standard
Dust (mg/m ³)	50 ¹ , 200 ²	20 ³ , 30 ⁴
SO ₂ (mg/m ³)	400 ¹ , 1200 ²	50 ³ , 200 ⁴
NOx (mg/m ³)	450 ¹ , 1100 ²	100 ³ , 200 ⁴

2004: ¹ New Power Plant ² Old Power Plant

2012: ³ Key Regions ⁴ Other Regions



Investment Cost

The ultra-low emission retrofit for a single 600MW unit costs about 150-250 million CNY.



Life Cycle Cost

Includes SCR catalyst replacement, desulfurizer consumption, and station service power.



Economic & Social Benefits

Intelligent retrofitting can reduce environmental chemicals consumption by 8%-15%.

DUAL CARBON GOALS - CARBON PEAKING AND CARBON NEUTRALITY

“30·60” Carbon Target

Carbon peaking by 2030, carbon neutrality by 2060



2021–2030: Achieve carbon peaking

2031–2045: Rapidly reduce carbon emissions

2046–2060: Deep decarbonization and carbon neutrality

- Clean and efficient utilization of coal in the power industry
- Vigorously develop new and renewable energy sources
- Carbon-reduction retrofits in key industries and fields
- Carbon capture, utilization, and storage technology (CCUS)

CCUS TECHNOLOGY – PRINCIPLE AND PATH COMPARISON

CCUS (Carbon Capture, Utilization, and Storage) is a core technical path to achieve deep decarbonization of coal power. By separating carbon dioxide from industrial emission sources and utilizing or storing CO₂, greenhouse gas emissions are greatly reduced.



Post-Combustion Capture

This technology is the most mature. It directly separates CO₂ from flue gas, with a simple process and easy transformation of active plants.



Pre-Combustion Capture

First, gasify fuels to produce syngas, then generate high-concentration CO₂ (30-60%) and hydrogen, reducing CO₂ separation energy consumption.



Oxy-Fuel Combustion

Use pure oxygen for combustion. Flue gas is mainly CO₂ and vapor, yielding high-concentration CO₂ (>90%) after condensation.

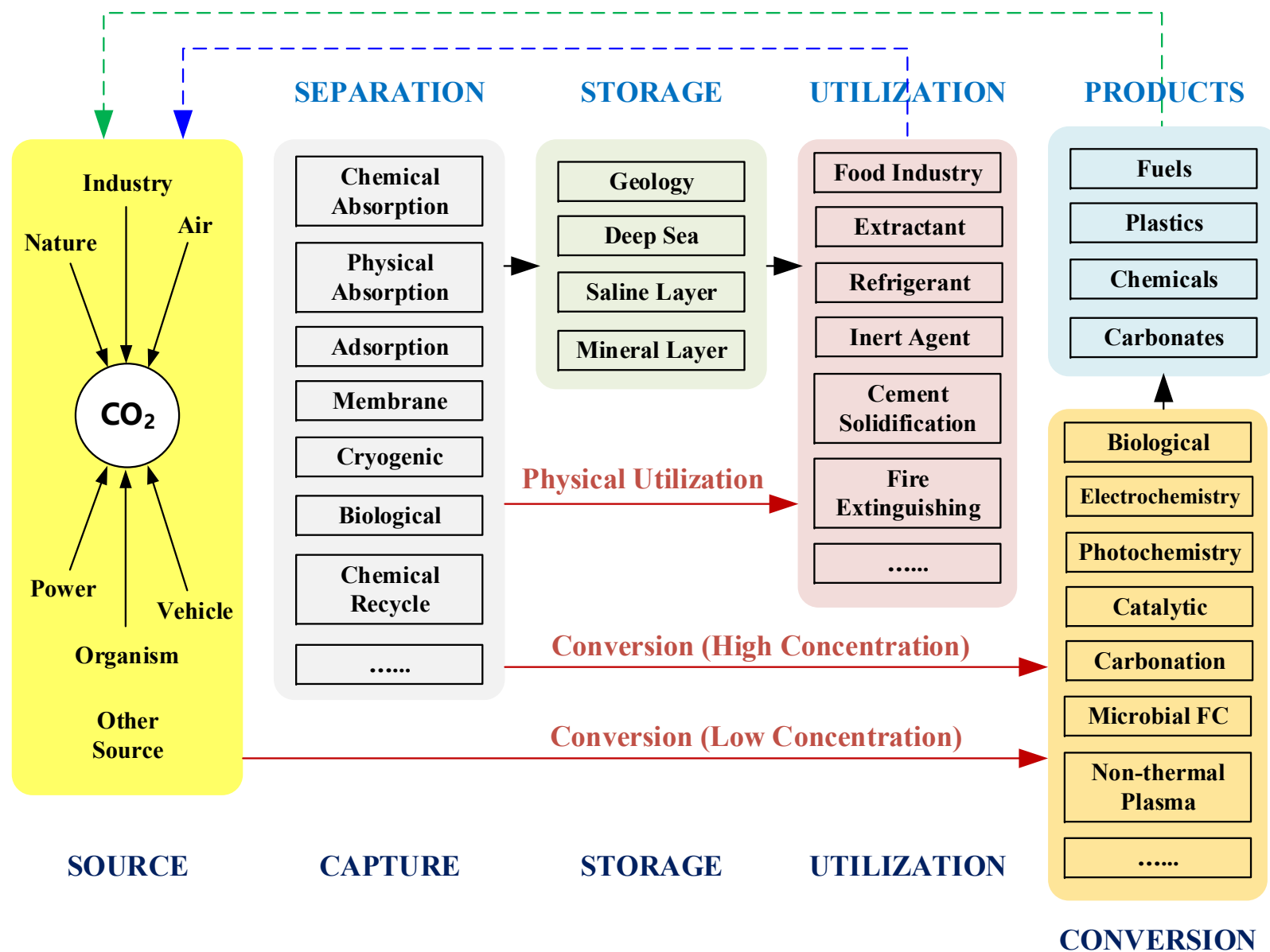
Core Insight: Existing units prioritize "Post-Combustion Capture" for retrofitting, new units can adopt "Pre-Combustion Capture" with IGCC, and "Oxy-Fuel Combustion" is a key direction for future upgrades.

CCUS

Full-Process

Technical

Framework



ULTRA-LOW EMISSION AND CCUS TECHNOLOGY



Ultra-Low Emission Technology Standards

Particulates $<5\text{mg}/\text{m}^3$, SO_2 $<35\text{mg}/\text{m}^3$, NO_x $<50\text{mg}/\text{m}^3$, even exceed emission standard of natural gas power plants.



World-Class Environmental Upgrade of China's Coal-fired Power Units

By the end of 2024, over 1050 GW of coal-fired power units in China had been retrofitted, accounting for over 94% of in-service units.



CCUS: The Key Technology Path for Deep Decarbonization



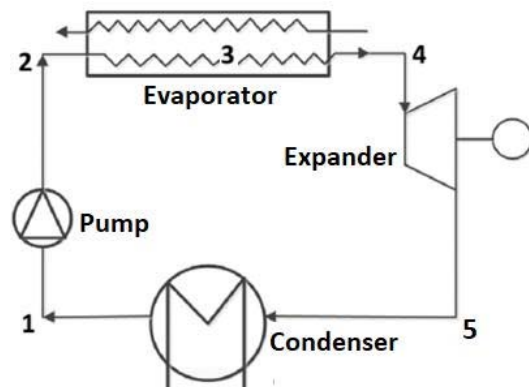
Benchmark Case: Zhengning Power Plant's megaton CCUS project, operated in 2025 with an annual capture capacity of 1.5 million tons.

ORGANIC RANKINE CYCLE (ORC) - PRINCIPLE AND ADVANTAGES

Through phase change cycle of organic working fluids, ORC can convert waste low-grade heat energy in industrial production into electrical energy, effectively improving overall energy efficiency level of power plant.

System Core Principle

The ORC system uses low-boiling organic working fluid, which absorbs low-temperature waste heat in evaporator to evaporate into high-pressure steam, drives expander to do work and generate electricity.



Core Technical Advantages



Low-Temperature Adaptability, High Energy Utilization Efficiency

It can efficiently utilize low-grade waste heat resources of 90-400°C, greatly improving the comprehensive energy recovery rate.



Safe and Reliable System, Convenient Operation and Maintenance

The system operates at low pressure, with compact equipment structure and high automation level, without complex operation.

ORC - PERFORMANCE COMPARISON AND APPLICATION

Performance Comparison with Steam Rankine Cycle

	Organic Cycle	Steam Cycle
Applicable Heat Source	Low/medium temp (90-400°C), wide range	High temp (>400°C), limited to high-grade
System Characteristics	Low pressure, compact, modular	High pressure, large area occupied
Resource Consumption	No water consumption, air cooling available	High water consumption, need water supply

Typical Application Scenario and Cases



Industrial Waste Heat Recovery: Flue Gas & Circulating Water

Install ORC systems at power plant tail end to recover wasted heat from low-temp flue gas and circulating water.



Marine Engineering: CNOOC Wenchang Oilfield

5MW ORC devices to recover waste heat from oil-field associated gas. Total power generation exceeds 10 million kWh.

COAL POWER'S INTEGRATION WITH RENEWABLE ENERGY

Connotation: Leverages the flexibility and regulating capacity of coal power to absorb intermittent wind and solar power, a key to building new power systems.

**Wind-Solar-Coal-Storage
Integration**

**Coal Power Flexibility
Retrofit**

**Cogeneration & Multi-
Energy Complementarity**

**Green Hydrogen Coupling
Combustion**



Strategic Outlook for Future Development

Coal power will transform from baseload source to "stabilizer" and "regulator".

Benchmark Practices and Future Trends

SPIC Shanghai Power Plant: USC units and biomass co-firing for low-carbon transition.

CHN Energy Datong Power Plant: 2000 MW wind-solar-coal-storage integrated project.

FOUR

CONCLUSION AND SUGGESTIONS



04 CONCLUSION AND SUGGESTIONS



Cost and Investment

Newly Built Units:

The investment in ultra-supercritical units is about 4,000 CNY/kW, with the best economic efficiency.

Retrofit Projects:

The flexibility retrofit cost is 30-90 CNY/kW; the CCUS retrofit cost is the highest, highly dependent on policy incentives.



Policy Incentive System

Existing System:

Mandatory standards, financial support, green finance, and market-based mechanisms.

Future Improvement:

Improve the capacity price mechanism, improve the ancillary service market, strengthen carbon market incentives.



Strategic Action Guidelines

- Deepen transformation and upgrading
- Focus on core technologies
- Promote energy hub transformation
- Stimulate endogenous motivation
- Strengthen international cooperation

04 INDUSTRY HOTSPOTS AND FUTURE OUTLOOK

Demand for Flexibility Enhancement

Deep peak shaving and rapid start-stop retrofitting of coal-fired units have become an industry hotspot.

Decarbonization and CCUS Integration

Integrating CCUS technology with traditional coal-fired power production processes is a key path for deep decarbonization.

Multi-Energy Integrated Energy Services

Coal power are transforming into integrated energy service including power, heat, cold, storage, green hydrogen, etc.

Intelligent and Digital Transformation

AI-optimized technologies improve operational efficiency and reduce maintenance costs, greatly reduce LCC.





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